

CURRICULUM VITAE

Aaron Kuang

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Citizenship: USA

EDUCATION

University of California, Irvine (UCI)

B.S. in Computer Science

Campuswide Honors Collegium, ICS Honors Program.

Honors Thesis: Help-seeking Behaviors in Algorithms Courses

Mar 2026

GPA: 4.00/4.00

RESEARCH INTERESTS

Theoretical Computer Science Education

I am interested in discovering and alleviating barriers to success in theoretical computer science courses. I'm curious about factors that influence sense of belonging, self-efficacy, and impostor syndrome in theoretical computer science. My goal is to evaluate the effectiveness of our current pedagogy and design interventions that facilitate student success.

Algorithm Design

I am interested in reductions and the application of subroutines as building blocks for larger algorithms. Formulating problems in ways that make existing subroutines useful opens up new ways to design efficient algorithms, while simultaneously simplifying the analysis to achieve good theoretical bounds and experimental performance.

RESEARCH EXPERIENCE

Help-seeking Behaviors in Algorithms Courses

Sep 2025 – Present

Advisor: Prof. Michael Shindler

Co-PI/Mentor: Prof. Stephen Tsung-Han Sher

- Co-authored a research study to investigate the overall sentiments and experiences that guide the help-seeking behaviors of students enrolled in algorithms courses.
- Coordinated and scheduled 22 interviews between members of the research group and study participants by collecting interviewer and interviewee availability.
- Oversaw the anonymization and cleaning of student transcripts, then assisted with qualitative coding, phenomenographic analysis, and paper writing.
- Attended two weekly meetings to discuss prior literature on help resource usage, lay down expectations for the next meeting, debrief on current progress, and identify cross-cutting trends seen in the transcripts.

Resisting Humanity's Mechanization in *Fahrenheit 451*

Apr 2023 – Jun 2023

Mentor: Dr. Katherine Raymond Walsh

- Wrote an independent literary analysis research paper examining how the technological dystopia of *Fahrenheit 451* affects its citizens' daily lives.
- Discussed how anti-intellectualism and over-reliance on technology has eroded the characters' need to think critically, rendering them blind to the horrors of the world.

- Reflected on how modern feats of technology—social media, virtual reality, and artificial intelligence—can erase our individuality and humanity if left unchecked.

TEACHING EXPERIENCE

Learning Assistant (University of California, Irvine):

Design and Analysis of Algorithms

Spring 2025, Fall 2025

Data Structure Implementation and Analysis

Fall 2024

AWARDS, HONORS, AND SCHOLARSHIPS

Outstanding Contribution to Research by an ICS Undergraduate Jun 2026

- Nominated by a UCI faculty member for contributions to research.

UCI PBK Fall Book Award Nov 2023

- Awarded to UCI sophomores who demonstrate academic excellence.

Victoria Alegria Tracy Endowed Scholarship Sep 2023

- Awarded to four UCI undergraduates annually.

UROP Humanities Core Research Paper Award Nomination Jun 2023

- Nominated by Dr. Katherine Walsh for a “superior research paper.”

AFFILIATIONS

Phi Beta Kappa Since Jun 2026

- Nominated by UCI’s Mu chapter as part of the top 5% of graduating seniors.

UCI ICS Honors Program (ICSHP)

Sep 2025 – Mar 2026

UCI Campuswide Honors Collegium (CHC)

Sep 2022 – Mar 2026